

Chapter 5 – DevOps & CI/CD

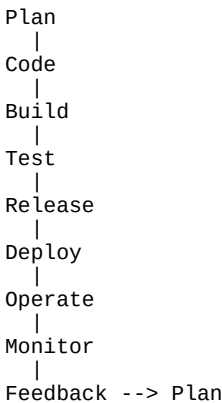
Learning Objectives

Understand DevOps culture, CI/CD concepts, DevOps lifecycle, tools, pipelines, best practices, and enterprise implementation.

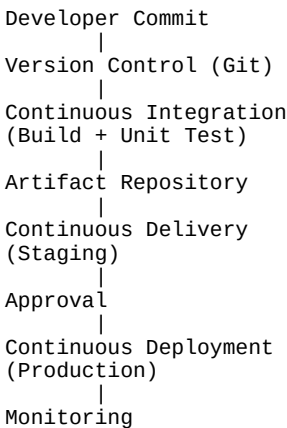
What is DevOps?

DevOps is a culture and engineering practice that unifies Development and Operations teams to automate software delivery, improve collaboration, and enable continuous delivery of high-quality software.

DevOps Lifecycle



CI/CD Pipeline



DevOps Toolchain

Stage	Popular Tools
Plan	Jira, Azure Boards
Code	Git, GitHub, GitLab, Bitbucket

Build	Maven, Gradle
CI/CD	Jenkins, GitHub Actions, Azure Pipelines
Test	JUnit, Mockito, Selenium
Container	Docker
Orchestration	Kubernetes
Monitoring	Prometheus, Grafana, ELK

Continuous Integration (CI)

Developers frequently merge code into a shared repository. Automated builds, unit tests, and quality checks validate every change, reducing integration issues.

Continuous Delivery (CD)

Every successful build is packaged and deployed automatically to staging or test environments, remaining ready for production release.

Continuous Deployment

Every validated change is automatically deployed to production without manual intervention, supported by monitoring and rollback strategies.

Benefits

- Faster releases
- Improved collaboration
- Better software quality
- Automated testing and deployment
- Reduced deployment risk
- Faster recovery from failures

Challenges

- Cultural change
- Tool integration
- Security and compliance
- Infrastructure automation complexity

Real-world Banking Example

A banking application uses Git for source control, Maven for builds, Jenkins for CI, JUnit and Selenium for automated testing, Docker for packaging, Kubernetes for deployment, and Prometheus/Grafana for monitoring. Each code commit triggers an automated pipeline.

Interview Questions

1. Explain DevOps.
2. CI vs Continuous Delivery vs Continuous Deployment.
3. Describe a CI/CD pipeline.

4. Why use Docker and Kubernetes?
5. How do rollback strategies work?
6. What are Infrastructure as Code and monitoring?